

AUTONOMOUS IT INFRASTRUCTURE OPERATIONS AND MANAGEMENT, POWERED BY AGENTIC AI

PHASE 1 REPORT

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BONAFIDE CERTIFICATE

Certified that this Project titled “**Autonomous IT Infrastructure Operations and Management, Powered by Agentic AI**” is the Bonafide work of **Lingeswaran V (230701163), Kumaran R (230701158), Miruthula KG (230701183), and Laksheta SV (230701161)**, who carried out the work under the guidance and supervision of Mr. M. Sudhakar, Associate Professor, Department of Computer Science and Engineering, Rajalakshmi Engineering College, Chennai.

Certified further that to the best of our knowledge, the work reported herein is based on the project development and study carried out by the above candidates and is submitted in partial fulfillment of the requirements for the award of the Bachelor of Engineering degree in Computer Science and Engineering.

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ABSTRACT

Autonomous IT Infrastructure Operations and Management, Powered by Agentic AI” is an intelligent IT operations assistant designed to support faster and context-aware investigation of infrastructure incidents. Modern enterprise environments combine networks, servers, databases, virtualization, containers, security controls, and multiple operational tools, making incident investigation increasingly complex and time-consuming.

The proposed system follows a local-first and supervised AI approach that combines a Small Language Model (SLM), Retrieval-Augmented Generation (RAG), agent-based task orchestration, memory, and LangChain-based tools. The SLM is intended to understand the incident, maintain task context, and plan the required investigation. RAG is intended to retrieve organization-specific information such as runbooks, infrastructure documentation, configuration details, and previous operational knowledge.

LangChain tools provide a structured interface between the AI agent and approved infrastructure capabilities. The system is designed to support networking, firewall operations, SSH-based system access, and other infrastructure diagnostics through modular tools. Human-in-the-loop approval is included as a control mechanism for sensitive or consequential actions, while secure service accounts and controlled access are intended to prevent unrestricted infrastructure access.

The current implementation includes the agent, agent memory, firewall operations, basic network tools, SSH connectivity, RAG-based knowledge retrieval, Human-in-the-Loop approval, database integration, and the frontend. The SLM has been developed and trained separately and is evaluated as the reasoning component of the system. The implemented system provides a complete foundation for supervised AI-assisted IT infrastructure operations, with controlled tool execution and administrator oversight for sensitive actions. The project focuses on secure and context-aware infrastructure assistance rather than unrestricted autonomous production deployment. The overall objective is to reduce repetitive manual investigation, improve the context available to engineers, and provide controlled AI assistance while preserving infrastructure security, privacy, and administrator authority.

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SIGNATURE OF THE MENTOR

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INTRODUCTION

1.1 GENERAL

Autonomous IT Infrastructure Operations and Management, Powered by Agentic AI” is an intelligent and supervised approach for assisting enterprise IT teams in investigating infrastructure incidents. Modern IT environments contain interconnected networks, servers, databases, applications, virtualization platforms, containers, security systems, and cloud or on-premise resources. As the number of components and dependencies increases, identifying the source of an incident requires information to be collected and correlated from several operational systems.

Traditional IT operations commonly use IT Service Management (ITSM), monitoring and observability systems, logging platforms, configuration data, knowledge bases, runbooks, and infrastructure administration tools. These systems provide the required information, but engineers often need to move between tools, interpret different forms of evidence, correlate events, and decide the next diagnostic step manually.

The proposed project introduces a local-first AI-powered IT Infrastructure Assistant that combines a Small Language Model (SLM), Retrieval-Augmented Generation (RAG), agent orchestration, memory, and LangChain-based tool integration. The SLM interprets the user's request and helps plan the investigation. RAG is designed to retrieve organization-specific information, while LangChain tools provide a structured and controlled way for agents to interact with approved infrastructure tools.

The system is intended to assist engineers rather than remove administrator authority. Basic investigation can be automated or assisted, while sensitive operations can be placed behind human approval and controlled credentials. This design aims to improve investigation speed and consistency while maintaining security, privacy, and operational control.

1.2 OBJECTIVES

- To develop a local-first AI-powered assistant for IT infrastructure investigation and operational support.
- To use an SLM to understand incident descriptions, maintain context, and coordinate investigation tasks.
- To integrate organization-specific knowledge through Retrieval-Augmented Generation (RAG).
- To provide controlled interaction with approved infrastructure tools using modular LangChain tools.
- To maintain context across interactions through agent memory.
- To support infrastructure diagnostics such as basic network checks, firewall operations, and SSH-based system access.
- To introduce administrator approval for sensitive or consequential infrastructure actions.
- To reduce repetitive manual troubleshooting and improve the context available during incident investigation.
- To provide a foundation that can be extended with routing, DNS, account vault, ITSM, observability, and additional infrastructure integrations.

1.3 EXISTING SYSTEM

Enterprise IT environments are generally managed through a combination of ITSM platforms, monitoring and observability systems, logging platforms, configuration management databases, knowledge bases, runbooks, and infrastructure administration tools. ITSM platforms record incidents, service requests, changes, and resolution history, while monitoring and logging systems provide operational evidence.

When an incident is reported, the support process generally begins with self-service or basic service desk investigation. L0 or self-service resources provide known solutions and guided troubleshooting. L1 service desk teams validate

impact, scope, affected users, and basic service health. If deeper technical analysis is required, the issue is escalated to L2 specialist teams such as application, database, network, or infrastructure teams. L3 engineering teams may become involved when source-code, architecture, or product-level changes are required.

Although the required tools and expertise are available, investigation often remains manual. Engineers search different systems, collect logs and metrics, compare recent changes, inspect configurations, and correlate evidence before deciding on the likely cause. The effectiveness of the process can therefore depend on engineer experience, documentation quality, and the information provided during escalation.

The existing approach is effective for structured IT operations, but it does not provide a single intelligent layer that can understand an incident, retrieve organization-specific knowledge, coordinate approved diagnostic tools, maintain investigation context, and present evidence-based next steps in one workflow.

1.4 PROBLEM STATEMENT

Modern enterprise IT infrastructure consists of interconnected networks, servers, applications, databases, firewalls, remote systems, and security components. As the number of infrastructure components increases, the process of identifying and resolving incidents becomes increasingly complex. A single incident may involve multiple systems and may require information to be collected from different operational sources before the actual cause can be identified.

Traditional infrastructure troubleshooting depends heavily on manual investigation. When an incident is reported, IT personnel generally examine system status, network connectivity, firewall configurations, remote machines, logs, documentation, and previous incident information. The engineer must then correlate these observations and determine the appropriate troubleshooting procedure. This process can become time-consuming,

particularly when the available information is distributed across different systems.

Another challenge is the repetitive nature of many infrastructure operations. Basic network diagnostics, firewall inspection, remote system checks, and information retrieval are frequently performed during incident investigation. Although these operations may be technically simple, performing them repeatedly increases the workload of IT personnel and can delay the overall investigation process.

The use of AI in infrastructure operations introduces another challenge. A language model can understand natural-language requests and generate troubleshooting suggestions, but allowing a model to directly execute infrastructure commands can create security and reliability risks. Incorrect tool selection, inappropriate commands, exposed credentials, or unintended configuration changes may affect the availability and security of enterprise systems.

Therefore, there is a need for an intelligent infrastructure assistant that can understand operational requests, maintain relevant context, retrieve organization-specific knowledge, and interact with infrastructure through controlled tools. The system should assist administrators without removing their authority over sensitive operations. This requirement motivates the development of a supervised AI-powered IT infrastructure operations system using a Small Language Model, agent orchestration, memory, Retrieval-Augmented Generation, LangChain tools, credential management, and Human-in-the-Loop control.

1.5 MOTIVATION OF THE PROJECT

The increasing complexity of enterprise infrastructure is one of the primary motivations for developing the proposed system. Modern organizations operate infrastructure consisting of multiple interconnected services and technologies. Network devices, servers, databases, security systems, applications, and remote machines may all contribute to the functioning of a

single business service. An incident affecting one component can therefore require investigation across several infrastructure layers.

A major motivation is to reduce the amount of repetitive manual work involved in infrastructure investigation. Operations such as checking network connectivity, examining firewall information, connecting to remote systems, and retrieving technical documentation are commonly performed by administrators during troubleshooting. An intelligent assistant can help coordinate these activities and present the collected information in a more understandable form.

Another motivation is to provide infrastructure assistance through natural-language interaction. Instead of requiring users to remember different commands or manually navigate multiple administration interfaces, an administrator can describe the problem in natural language. The agent can interpret the request and identify the appropriate capability for investigation.

The project is also motivated by the need for **context-aware troubleshooting**. An infrastructure investigation is rarely limited to a single question. Administrators may ask follow-up questions based on previous observations. Maintaining relevant conversational context allows the assistant to support a continuous investigation rather than treating every request as an independent operation.

The availability of organization-specific knowledge is another important motivation. General-purpose language models may not contain information about an organization's internal architecture, runbooks, configurations, or operational procedures. Retrieval-Augmented Generation provides a mechanism for incorporating such information into the investigation process.

Finally, enterprise infrastructure requires strong security and operational control. The proposed system therefore follows a supervised approach in which infrastructure access is provided through controlled LangChain tools and sensitive operations can be subjected to administrator approval. This

allows AI assistance to be introduced without giving the model unrestricted authority over enterprise infrastructure.

1.6 SCOPE OF THE PROJECT

The scope of the proposed project is the development of an AI-assisted infrastructure operations system capable of supporting investigation and controlled interaction with IT infrastructure.

The system covers the following major areas:

- Natural-language interaction through a web-based frontend.
- Agent-based coordination of infrastructure investigation tasks.
- Agent memory for maintaining relevant conversational context.
- A separately developed Small Language Model for infrastructure-oriented language understanding and reasoning.
- Retrieval-Augmented Generation for accessing organization-specific knowledge.
- LangChain-based tools for controlled interaction with infrastructure capabilities.
- Basic network diagnostics and connectivity investigation.
- Firewall-related monitoring and operations.
- SSH-based remote system connectivity.
- Routing and DNS-related infrastructure investigation.
- Credential management for controlled access to infrastructure systems.
- Database support for storing relevant operational information.
- Human-in-the-Loop control for sensitive or consequential operations.
- Docker-based deployment and testing of the infrastructure workflow.

The project focuses on **AI-assisted and supervised infrastructure operations** rather than unrestricted autonomous control. The system is intended to support administrators by reducing repetitive investigation work, collecting relevant information, and providing structured recommendations.

The SLM and operational agent are maintained as separate components in the current implementation. The SLM can be developed and evaluated

independently, while the agent provides orchestration and interaction with available infrastructure tools. Complete integration of the separately developed SLM into the operational agent workflow can be considered as a future enhancement.

The project does not aim to replace IT administrators or completely automate enterprise infrastructure management. Instead, it establishes a controlled foundation that can be extended with additional infrastructure capabilities and enterprise integrations.

1.7 SIGNIFICANCE OF THE PROJECT

The proposed system is significant because it combines AI-based reasoning with controlled infrastructure operations in a single assistance framework. Traditional IT operations generally require administrators to switch between multiple tools, interpret technical information, and manually determine the next investigation step. The proposed assistant provides an additional intelligent layer that can help coordinate these activities.

For **IT administrators**, the system can reduce repetitive diagnostic work by providing natural-language access to supported infrastructure capabilities. Network checks, firewall information, remote-system investigation, and knowledge retrieval can be coordinated through the agent.

For **IT support and operations teams**, the system can help improve the consistency of incident investigation. The assistant can maintain context, retrieve relevant organizational information, and provide structured findings instead of requiring every step to be performed manually.

The use of **Retrieval-Augmented Generation** is significant because it allows organization-specific information to be incorporated into the assistance process. Internal documentation, operational knowledge, and technical resources can provide context that would not necessarily be available within the language model itself.

The use of **LangChain tools** provides an important separation between AI reasoning and infrastructure execution. Instead of providing unrestricted access to the underlying systems, infrastructure capabilities can be exposed through defined tools. This provides a more controlled mechanism for the agent to interact with infrastructure.

Security and administrator control are also important aspects of the project. Credential management and Human-in-the-Loop mechanisms provide additional safeguards for infrastructure operations. Sensitive actions can be subjected to administrator approval, ensuring that AI assistance does not automatically override operational authority.

Overall, the project demonstrates how Small Language Models, agent-based orchestration, memory, RAG, controlled tools, and human supervision can be combined to create a practical foundation for intelligent IT infrastructure operations.

CHAPTER 2

LITERATURE SURVEY

2.1 AI for IT Operations and AIOps

AIOps applies artificial intelligence and machine learning techniques to IT operational data such as metrics, logs, alerts, and events to support anomaly detection, incident analysis, failure management, and operational efficiency. The increasing scale and heterogeneity of infrastructure environments make manual analysis difficult, creating a need for intelligent systems that can process operational information and assist administrators.

Zhang et al. (2024) surveyed the application of Large Language Models to AIOps and discussed their potential across areas such as anomaly detection, diagnosis, and failure management. These developments demonstrate the growing role of AI in supporting modern IT operations.

2.2 Language Models for IT Assistance

Language models can assist IT engineers by interpreting incident descriptions, summarizing operational information, generating troubleshooting guidance, and interacting with users through natural language. Research on reasoning and tool-using language models has demonstrated how models can combine reasoning with external actions.

Yao et al. (2023) introduced ReAct, which integrates reasoning and action for interaction with external environments, while Schick et al. (2023) explored the use of external tools through Toolformer. These approaches provide a foundation for developing AI assistants capable of supporting operational tasks beyond simple text generation

2.3 Small Language Models for Local Deployment

Small Language Models provide a practical option for applications where local execution, lower computational requirements, privacy, and controlled deployment are important. Minaee et al. (2024) surveyed Small Language

Models and their applications, highlighting their potential for efficient deployment in resource-constrained environments.

A local SLM can therefore be used as a planning and reasoning component while access to infrastructure resources is maintained through separately controlled tools and services. This approach is particularly relevant for enterprise environments where operational information may need to remain within organizational boundaries.

2.4 Retrieval-Augmented Generation

Retrieval-Augmented Generation combines language-model generation with information retrieved from external knowledge sources. Lewis et al. (2020) introduced the RAG approach for knowledge-intensive tasks by combining parametric language-model knowledge with retrieved information.

In IT operations, retrieved information can include runbooks, configuration documentation, architecture information, previous incident records, and operational manuals. The effectiveness of such a system depends on factors including the quality of the source documents, document chunking, indexing, metadata, access permissions, and information freshness.

2.5 Agent-Based AI Systems

LangChain provides a structured framework for connecting AI agents with external tools and infrastructure capabilities. Within an infrastructure environment, individual LangChain tools can expose controlled capabilities for activities such as network diagnostics, SSH operations, firewall management, and system administration.

This separation allows the agent to select and invoke specific tools without requiring unrestricted direct access to the underlying infrastructure. The proposed system uses LangChain tools to organize infrastructure capabilities as controlled operations that can be invoked according to the user's request and the defined operational permissions.

2.6 LangChain Tools for Infrastructure Operations

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2.7 Human-in-the-Loop Automation

Infrastructure automation must distinguish between read-only diagnostic operations and actions that modify system state. Human-in-the-loop approaches provide administrator oversight before sensitive operations are executed. D'Antonio and Xie (2025) examined human-in-the-loop runbook improvement and supervised support automation, emphasizing the role of human operators in operational workflows. In an infrastructure assistant, approval mechanisms can therefore be used before actions such as configuration changes, firewall modifications, service operations, or other potentially disruptive activities.

2.8 Security and Privacy

Enterprise AI systems must address security and privacy requirements including credential protection, least-privilege access, auditability, data exposure, and controlled deployment. A local-first architecture can reduce dependence on external services when sensitive infrastructure information is involved.

Separating the language model from infrastructure credentials and providing access through controlled tools can further restrict the authority available to

the AI system. The proposed architecture incorporates these principles through controlled tool execution, credential management, and administrator approval for sensitive operations

2.9 Research Gap

Existing research demonstrates the value of AI for operational analytics, language assistance, retrieval-augmented generation, tool use, and automated incident management. However, these capabilities need to be integrated carefully when an AI system is connected to heterogeneous IT infrastructure.

The proposed project addresses this integration by combining a Small Language Model, Retrieval-Augmented Generation, memory, agent-based orchestration, LangChain tools, and Human-in-the-Loop supervision within a local-first infrastructure operations framework. This provides a controlled approach in which the AI can assist with infrastructure tasks while administrator authority remains central to sensitive operations.

CHAPTER 3

PROPOSED SYSTEM

3.1 GENERAL

The proposed system is a local-first AI-powered IT Infrastructure Assistant designed to support incident investigation and controlled infrastructure operations. The system receives an incident or operational request through the frontend and uses an agent layer to understand the task, retain context, and determine the investigation steps.

The target Small Language Model for the project is Qwen 8B. It is intended to interpret natural-language infrastructure requests and assist in planning the required actions. Organization-specific information is intended to be supplied through a RAG pipeline so responses can use relevant internal documentation. Infrastructure interaction is separated from model reasoning through **LangChain-based tools**. This allows capabilities such as firewall operations, basic network diagnostics, and SSH access to be exposed as controlled tools. The design can later be extended with routing, DNS, database, container, observability, and ITSM integrations.

Human-in-the-loop control is planned for sensitive operations. This creates a boundary between investigation and consequential change: the assistant may gather evidence and recommend actions, while an administrator can approve operations that modify infrastructure state.

3.2 SYSTEM ARCHITECTURE

The system is organized into logical layers: **User Interface, Agent and Memory, SLM, RAG Knowledge, LangChain Tool Integration, and Security/Control**.

The User Interface receives incident descriptions and presents results. Agent and Memory coordinate tasks and preserve context. The SLM performs language understanding and planning. RAG retrieves organization-specific knowledge. **LangChain tools provide controlled infrastructure access**. Security and Control manages permissions, credentials, and approval boundaries

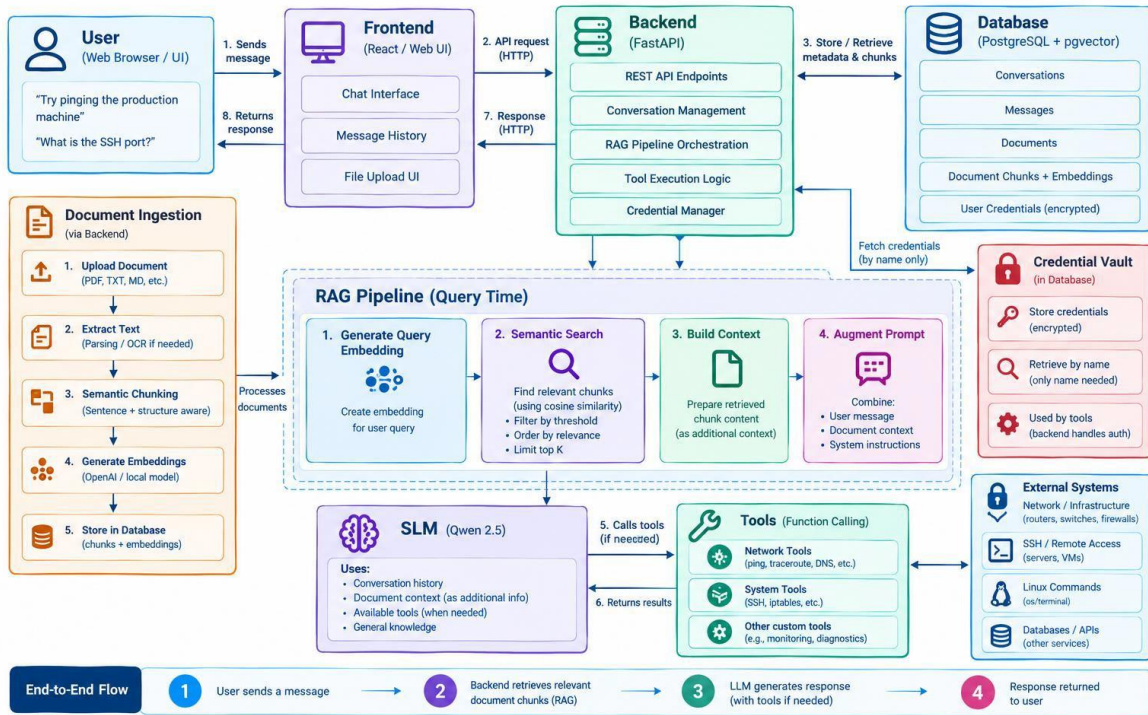


Fig 3.1: System Architecture

LAYER	MAJOR COMPONENTS	PURPOSE
User Layer	Frontend	Accepts requests and displays findings and recommendations.
Agent Layer	Agent, Memory	Coordinates investigation tasks and preserves context.
AI Layer	Qwen 8B SLM	Interprets requests and supports diagnostic planning.
Knowledge Layer	RAG, internal documents	Retrieves relevant technical information.
Tool Layer	Langchain tools, Network, Firewall, SSH	Provides controlled infrastructure diagnostics and operations.
Control Layer	HITL, Account Vault, permissions	Protects sensitive actions and credentials.

3.3 DEVELOPMENTAL ENVIRONMENT

3.3.1 HARDWARE REQUIREMENTS

The hardware environment depends on the local deployment configuration and resources required by the selected SLM. Exact production requirements will depend on model quantization, workload, concurrent users, and integrated infrastructure tools.

COMPONENTS	SPECIFICATION / PURPOSE
Processor	Modern multi-core processor suitable for local AI development and tool execution

RAM	Adequate memory for SLM, retrieval services, application services, and development tools
Storage	SSD storage for models, documents, indexes, logs, and application data
Network	Internal connectivity to approved infrastructure systems

3.3.2 SOFTWARE REQUIREMENTS

COMPONENTS	SPECIFICATION
Operating System	Linux / Windows development environment
Programming Language	Python
Language Model	Qwen 8B SLM
AI Architecture	Agent Orchestration with langraph
Knowledge Retrieval	RAG and vector retrieval
Tool Integration	Langchain tools
Frontend	Web-based project frontend
Infrastructure Access	SSH, network and firewall tools
Database	Postgres and pgvector

3.4 DESIGN OF THE ENTIRE SYSTEM

3.4.1 SYSTEM OVERVIEW

The proposed system is designed as a modular AI-assisted infrastructure management framework. The system provides a structured interaction between the user, frontend, agent, reasoning model, knowledge retrieval layer, and infrastructure tools.

The workflow begins when the user submits an infrastructure-related request through the frontend. The agent interprets the request and examines the available context before determining the required course of action. When organization-specific information is needed, the RAG component retrieves relevant documentation and provides the required context to the agent.

Based on the request and retrieved information, the agent develops a diagnostic plan and selects the appropriate langchain tools. These tools interact with the infrastructure to collect relevant evidence, such as network status, firewall information, and observations from remote systems. The collected results are then returned to the agent for analysis.

For operations that may modify the infrastructure or change system state, the design incorporates a Human-in-the-Loop mechanism through which administrator approval can be requested before execution. After analysing the available evidence, the system presents the findings and recommended next action to the user.

Overall, the system design separates **reasoning**, **knowledge retrieval**, **tool execution**, and **administrative control**, providing a structured approach for AI-assisted IT infrastructure operations.

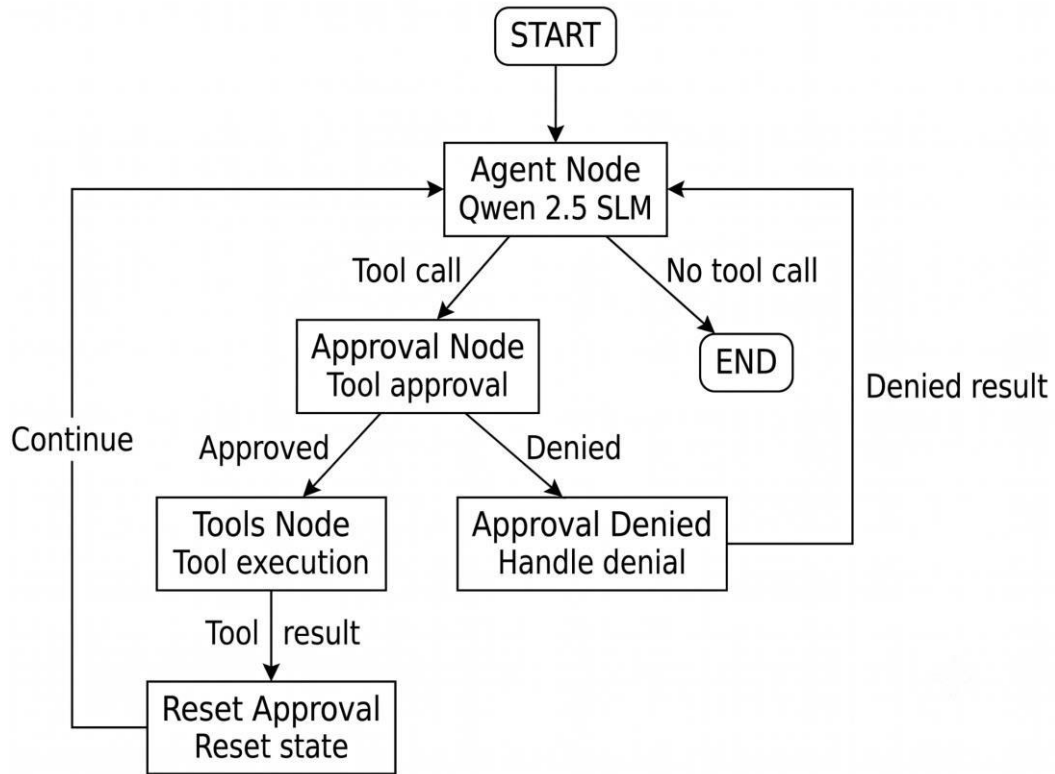


Fig 3.2: Human-in-the-Loop Approval Flow

STEP	ACTIVITY
1	User submits incident or infrastructure request
2	Agent interprets request and checks memory/context
3	Relevant organizational knowledge is retrieved
4	Agent creates a diagnostic plan
5	Approved lanegchain tools are invoked
6	Infrastructure evidence is collected
7	Evidence is correlated and likely cause is identified
8	Recommendation is generated; sensitive actions require approval
9	Result is displayed to the user

3.5 COMPARATIVE ANALYSIS

ASPECT	EXISTING APPROACH	PROPOSED SYSTEM	EXPECTED BENEFIT
Incident Investigation	Manual collection across tools	Agent-assisted workflow	Reduced repetitive effort
Knowledge Access	Manual search	RAG-based retrieval	Faster relevant information access
Tool Interaction	Manual commands/tool switching	Controlled MCP invocation	Structured tool use
Context	Spread across ticket/notes	Agent memory	Better continuity
Decision Support	Experience dependent	SLM-supported analysis	Improved guidance
Security Control	Enterprise permissions	Controlled tools + planned HITL	Administrator authority
Deployment	May depend on external services	Local-first architecture	Improved privacy and internal control

3.6 DESIGN PRINCIPLES

The proposed system is designed around a set of principles intended to make AI-assisted infrastructure operations practical, modular, secure, and extensible. These principles guide the interaction between the user interface, agent, Small Language Model, memory, RAG system, LangChain tools, infrastructure systems, and security components.

3.6.1 Local-First Operation

The system follows a local-first approach in which infrastructure-related processing can be performed within the controlled environment of the organization. This is particularly important because infrastructure environments may contain sensitive information such as server details, network configurations, internal documentation, and operational records.

3.6.2 Modular Architecture

Each major capability is implemented as a separate module. The system contains components for agent orchestration, memory, knowledge retrieval, credential management, infrastructure tools, database operations, and user interaction. This modular design allows individual components to be developed, tested, and improved independently.

3.6.3 Separation of Reasoning and Execution

The system separates AI reasoning from infrastructure execution. The agent interprets the user's request and determines the appropriate operation, while infrastructure interaction is performed through defined LangChain tools. This prevents the reasoning component from receiving unrestricted access to infrastructure systems.

3.6.4 Controlled Infrastructure Access

Infrastructure capabilities are exposed through specific tools rather than unrestricted system access. Each tool performs a defined operation and returns structured information to the agent. This provides a controlled boundary between the AI system and the infrastructure environment.

3.6.5 Context-Aware Assistance

The system maintains relevant information from previous interactions through agent memory. This allows follow-up questions to be interpreted using the context of the ongoing investigation rather than treating every request as an isolated interaction.

3.6.6 Knowledge-Grounded Responses

The RAG component allows the system to retrieve relevant organization-specific information when required. This enables the assistant to use available documentation and operational knowledge rather than depending entirely on information contained within the language model.

3.6.7 Human Supervision

The system maintains administrator authority over sensitive operations. Operations that may modify infrastructure state can be subjected to Human-in-the-Loop approval before execution. This provides an additional safety layer between AI recommendations and consequential infrastructure actions.

3.6.8 Least-Privilege Access

The system is designed around the principle that infrastructure access should be limited to the permissions required for a particular operation. Credential management and controlled tools help prevent unnecessary exposure of infrastructure privileges.

3.6.9 Auditability

Important system interactions can be recorded to support operational analysis and accountability. User requests, tool operations, retrieved information, approvals, and resulting actions can form part of the operational record.

3.6.10 Extensibility

The modular structure allows additional infrastructure capabilities to be added without redesigning the complete system. New LangChain tools, knowledge sources, infrastructure modules, and operational workflows can be introduced as the system evolves.

3.7 SECURITY ARCHITECTURE

Security is an essential part of an AI system that interacts with IT infrastructure. The proposed system therefore treats AI reasoning and infrastructure execution as separate stages. The agent can determine what needs to be investigated, but access to infrastructure is provided through controlled mechanisms.

The overall security-oriented interaction can be represented as:

User → Frontend → Agent → Permission and Tool Selection → LangChain Tool → Credential Management → Infrastructure → Result → Agent → User

The frontend provides the initial interaction point. The user submits an infrastructure-related request through the application. The request is passed to the agent, which interprets the requirement and determines whether additional knowledge or infrastructure information is required.

When an infrastructure operation is necessary, the agent selects an appropriate LangChain tool. The tool provides a defined interface for the required operation. Before access to a protected system is performed, the required credentials are obtained through the credential-management mechanism rather than being directly exposed through the natural-language interaction.

For sensitive operations, Human-in-the-Loop control provides an additional authorization stage. Instead of immediately executing an operation that may change infrastructure state, the system can request administrator confirmation. The operation proceeds only after the required approval is obtained.

3.7.1 Credential Protection

Infrastructure credentials are maintained separately from normal conversational interaction. This reduces the possibility of exposing sensitive authentication information in prompts or generated responses.

3.7.2 Access Control

Access to infrastructure is provided according to the capabilities required by the tool. This allows the system to avoid providing unrestricted access to every infrastructure component.

3.7.3 Tool-Level Control

Each LangChain tool represents a specific capability. Tool definitions establish what operation the agent can request and what information can be returned.

3.7.4 Human Approval

Sensitive operations can be placed behind an approval mechanism. This ensures that the AI assistant supports administrators rather than independently overriding administrative decisions.

3.7.5 Operational Logging

The system can maintain records of relevant requests and operations. These records support troubleshooting, auditing, and analysis of the assistant's behaviour.

3.8 SEPARATION OF SLM AND AGENT

The Small Language Model and the operational agent are treated as separate components in the current system. This separation allows the language model and infrastructure orchestration components to be developed and evaluated independently.

The **Small Language Model** is responsible for language understanding and reasoning-related capabilities. It processes natural-language information and provides the model-level intelligence required for infrastructure-oriented assistance.

The **Agent**, on the other hand, is responsible for coordinating the operational workflow. It receives the user's request, determines the required investigation steps, maintains context, accesses relevant knowledge, selects available tools, and processes the resulting observations.

The responsibilities can therefore be represented as:

Component	Primary Responsibility
Small Language Model	Language understanding and reasoning
Agent	Task orchestration and decision coordination
Agent Memory	Context retention
RAG	Organization-specific knowledge retrieval
LangChain Tools	Infrastructure interaction
Credential System	Controlled infrastructure authentication
Database	Operational data persistence
HITL	Administrator control

This separation is useful because model development and infrastructure operations have different requirements. The SLM can be trained and evaluated using appropriate datasets without directly connecting the model to infrastructure systems during the model-development stage.

Similarly, the agent can be tested independently using available tools and operational workflows. This reduces the coupling between model experimentation and infrastructure execution.

The current implementation therefore establishes the SLM and agent as distinct components. Complete end-to-end integration of the separately developed SLM into the operational agent workflow remains an area for future enhancement.

3.9 RETRIEVAL-AUGMENTED GENERATION ARCHITECTURE

Retrieval-Augmented Generation is used to provide the system with access to organization-specific knowledge. General language models may not contain information about an organization's internal infrastructure, procedures, architecture, or operational documentation. RAG addresses this limitation by retrieving relevant information from an external knowledge source during the interaction.

The implemented RAG workflow can be represented as:

Document Upload → Document Processing → Knowledge Indexing → User Query → Relevant Information Retrieval → Context Formation → AI Response

The process begins when organization-specific documents are added to the knowledge base. These documents may contain technical documentation, operational procedures, runbooks, configuration information, or other approved sources of infrastructure knowledge.

The documents are processed and prepared for retrieval. Relevant content is indexed so that it can be searched when a user submits a query.

When the user asks a question that requires organization-specific information, the system identifies relevant knowledge from the indexed information. The retrieved information is then provided as context for generating the response.

This approach reduces the dependency on information stored solely within the language model. It also allows the knowledge base to be updated independently as organizational documentation changes.

3.9.1 Knowledge Sources

Potential knowledge sources include:

- Infrastructure documentation
- Operational runbooks
- Troubleshooting procedures
- Architecture documentation
- Configuration references
- Previous incident knowledge
- Approved technical documentation

3.9.2 Knowledge Retrieval

The retrieval process attempts to identify information relevant to the user's current request. Only relevant information needs to be supplied to the reasoning process rather than loading the complete knowledge base for every interaction.

3.9.3 Benefits of RAG

RAG provides several advantages:

- Access to organization-specific knowledge
- Reduced dependence on model memorization
- Easier knowledge updates
- Better contextual responses
- Support for internal documentation
- Improved consistency with organizational procedures

3.10 DATABASE DESIGN

The database provides persistent storage for information required by the application and its operational workflow. While conversational reasoning may occur dynamically, important application information must be stored in a structured manner for later retrieval and management.

The database can support information associated with users, sessions, infrastructure interactions, tool executions, knowledge resources, approvals, and operational records.

3.10.1 User and Session Information

User-related information can be associated with application sessions and interactions. This allows the system to maintain a structured relationship between users and their operational activities.

3.10.2 Incident Information

Relevant incident information can be maintained as part of the operational record. This may include the reported issue, investigation status, observations, recommendations, and resolution information.

3.10.3 Tool Execution Records

Tool-related information can be recorded to provide visibility into which infrastructure capability was used during an investigation and what result was obtained.

3.10.4 Knowledge Metadata

Information associated with uploaded knowledge resources can be maintained to support knowledge-base management and retrieval.

3.10.5 Approval Records

Human approval activities can be associated with sensitive operations. This provides a record of when an operation required administrator authorization and whether the operation was approved.

3.10.6 Audit Information

Operational records can support later investigation of system activity. This is particularly important for enterprise environments where infrastructure operations need to remain traceable.

CHAPTER 4

MODULE DESCRIPTION

4.1 SYSTEM ARCHITECTURE

The proposed system is divided into modular components so reasoning, memory, knowledge retrieval, infrastructure interaction, and user interaction can be developed and controlled independently. The modular design also allows new infrastructure capabilities to be added without changing the core reasoning layer.

4.1.1 FRONTEND AND USER INTERACTION

The frontend is the primary interface through which an engineer submits an infrastructure issue or operational request. It is intended to display the assistant response, investigation status, relevant evidence, and recommended actions. The interface acts as the entry point to the agent workflow and does not directly expose unrestricted infrastructure credentials.

4.1.2 BACKEND AND AGENT INFRASTRUCTURE

The backend coordinates the agent, memory, model, retrieval, and tool layers. The agent receives the request and determines the investigation sequence. Memory stores relevant conversation or task context. The backend also provides the boundary through which MCP tools are invoked and their results returned.

4.2 DATA COLLECTION AND KNOWLEDGE PROCESSING

4.2.1 Infrastructure Data Collection

The system is intended to collect operational information from approved infrastructure sources. Examples include network diagnostics, firewall state, remote-system information obtained through SSH, logs, configuration information, and tool-specific evidence. Current development has focused on basic network tools, firewall functionality, and SSH connectivity.

4.2.2 Knowledge Sources

The planned RAG knowledge base can contain organization-specific technical documentation such as network diagrams, server configurations, database schemas, operational manuals, runbooks, troubleshooting procedures, and relevant incident records.

4.2.3 Retrieval Process

The planned RAG workflow converts source documents into searchable representations, retrieves information relevant to the current incident, and supplies the retrieved context to the SLM. Retrieval should respect source permissions and preserve metadata identifying the information source.

4.2.4 Model and Agent Strategy

Qwen 8B is selected as the target SLM. The model is used as a reasoning and planning component rather than an unrestricted infrastructure executor. The model is trained.

4.3 SYSTEM WORKFLOW

4.3.1 User Interaction

The user enters an infrastructure incident or operational request through the frontend.

4.3.2 Request Understanding

The agent interprets the request, identifies the affected service or component, and checks memory/context.

4.3.3 Knowledge Retrieval

When implemented, RAG retrieves relevant organization-specific documentation and previous operational knowledge.

4.3.4 Diagnostic Planning

The agent creates a sequence of investigation steps based on the issue and available evidence.

4.3.5 Tool Interaction

Approved Langchain tools perform diagnostics such as network checks, firewall inspection, and SSH-based access.

4.3.6 Evidence Analysis

Returned results are correlated to identify likely causes and the next diagnostic step.

4.3.7 Human Validation

Sensitive state-changing actions are intended to be placed behind administrator approval.

4.3.8 Result Display

The frontend presents findings, evidence, and recommended actions.

4.4 MODULE SUMMARY

MODULE	DESCRIPTION
Agent	Understands requests and coordinates required tasks.
Memory	Maintains context from previous interactions.
Network Tools	Performs basic network diagnostics and checks.
Firewall	Handles firewall monitoring and operations.
SSH	Enables secure access to remote systems.
Routing	Supports routing analysis and troubleshooting; under development.

DNS	Supports DNS resolution and troubleshooting; under development.
RAG	Retrieves relevant organization-specific knowledge;
Routing	Routing Environment and tools are under development.
Account Vault	Securely manages infrastructure credentials; planned.
Human-in-the-Loop	Enables administrator approval for sensitive actions; under development.
Frontend	Provides the interface for user interaction.
Model & Database	Supports model-related data collection and information storage; database schema under development.
Integration	Connects the modules into a unified workflow; under development.

CHAPTER 5

IMPLEMENTATION AND RESULTS

5.1 IMPLEMENTATION

The proposed system is implemented as a modular, local-first AI infrastructure assistant designed to support IT infrastructure investigation and management. The system combines a web-based frontend, Python-based backend services, a Small Language Model (SLM), an agent layer, agent memory, and infrastructure tools.

The frontend provides an interface through which users can submit infrastructure-related requests in natural language. The backend processes these requests and communicates with the agent layer. The agent works with the SLM to understand the request, determine the required action, and select the appropriate infrastructure tool.

The implementation includes agent memory, basic network tools, firewall functionality, SSH connectivity, and a web-based user interface. Docker is used to provide an isolated environment for the system components and to support communication between the agent and target infrastructure.

The system follows a controlled execution approach in which the SLM is responsible for reasoning and planning, while infrastructure operations are performed through separate tools. This separation prevents the language model from directly executing unrestricted infrastructure operations.

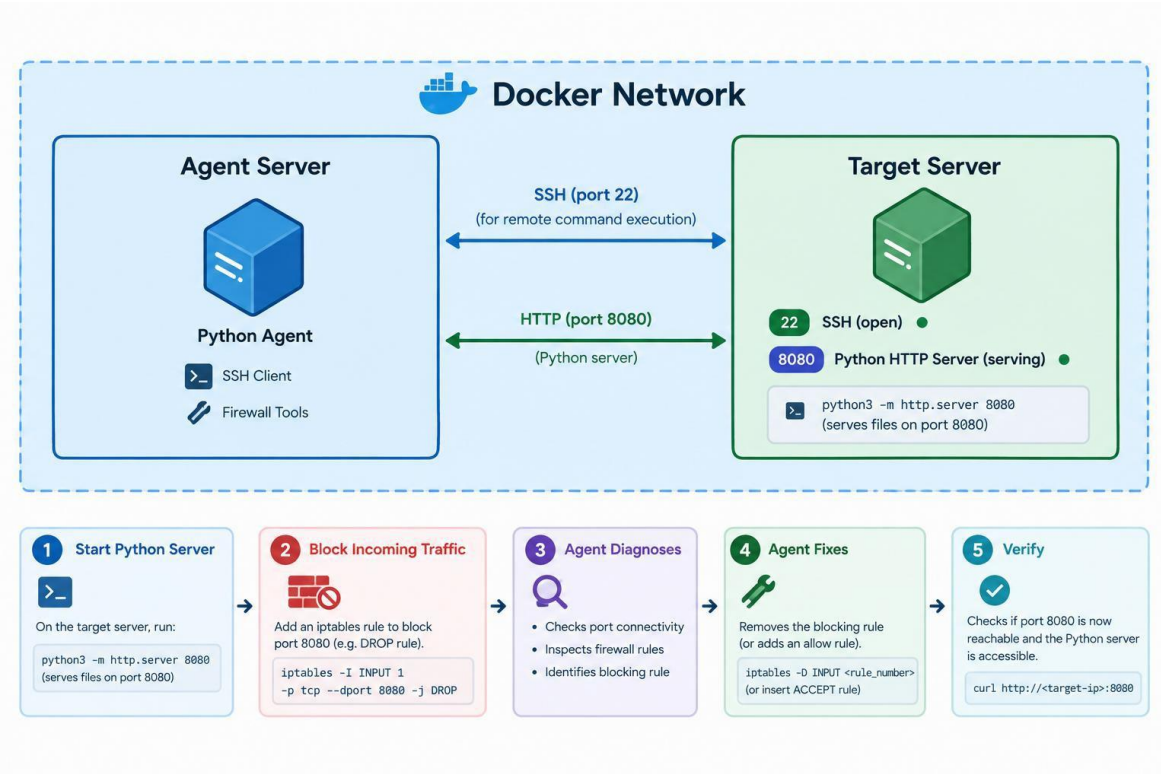


Fig 5.1: Docker Network and Remote Execution Workflow

5.1.1 Agent Implementation

The agent acts as the central coordination component of the system. It receives the user's request and works with the SLM to understand the intent and determine the appropriate course of action.

Based on the request, the agent can select suitable infrastructure tools such as network diagnostics, SSH, and firewall-related operations. The selected tool performs the required operation and returns its result to the agent. The agent then interprets the result and provides the appropriate response to the user.

Agent memory is incorporated to maintain relevant information across interactions. This enables the system to use previous context while processing follow-up requests.

The agent workflow includes request interpretation, reasoning, tool selection, tool execution, result processing, and response generation. Sensitive infrastructure operations can be controlled through a Human-in-the-Loop approval mechanism, allowing an administrator to approve or deny actions before execution.

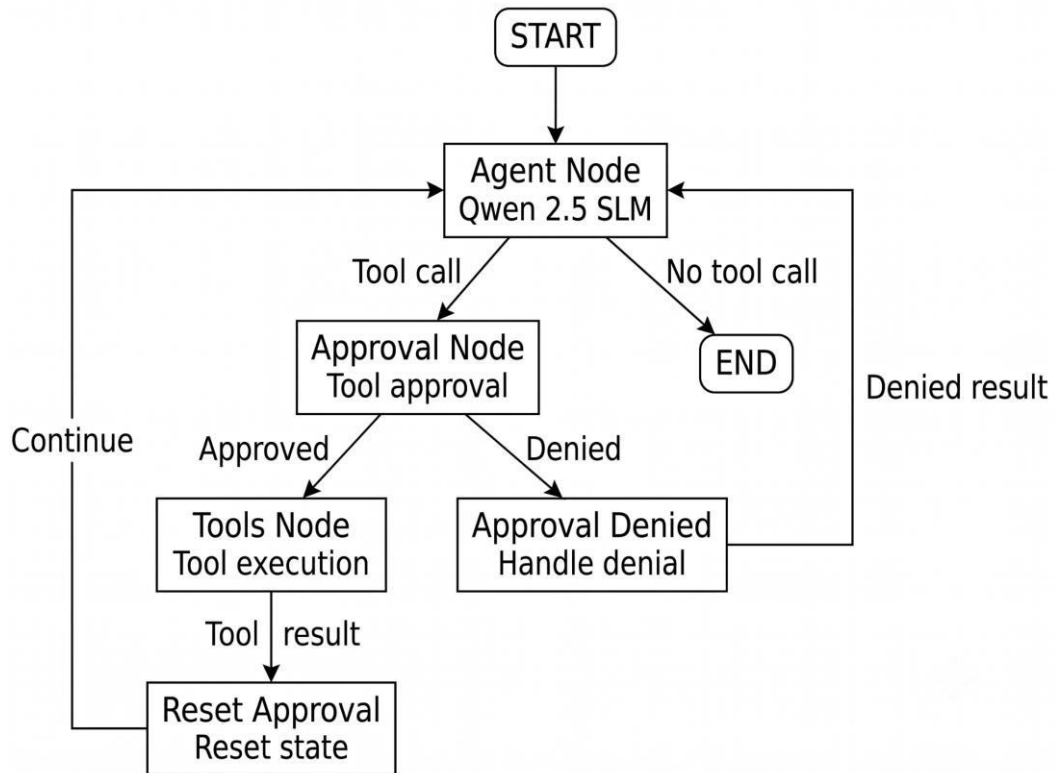


Fig 5.2: Agent Workflow with Tool Approval

5.1.2 SLM Development

The Small Language Model is developed as a separate reasoning component of the system. The SLM is trained using infrastructure-oriented data and is designed to understand IT-related requests and support agent decision-making.

The SLM development process includes dataset preparation, preprocessing, model training or fine-tuning, evaluation, and integration with the agent. The trained model is evaluated separately before being connected to the infrastructure execution layer.

The SLM provides reasoning and planning capabilities, while actual infrastructure operations are performed through controlled tools. This separation provides better control over infrastructure access and reduces the possibility of unrestricted model-generated commands being executed directly.

5.1.3 Infrastructure Tool Implementation

The infrastructure tool layer provides controlled access to IT infrastructure operations. The implemented tools include basic network diagnostic capabilities, SSH-based remote access, and firewall-related operations.

Network tools can be used to investigate connectivity and network-related issues. SSH enables the agent to communicate with remote systems and perform approved diagnostic operations. Firewall tools support the inspection and management of firewall-related configurations.

The tool layer separates infrastructure execution from the reasoning process of the SLM. This modular approach allows additional tools such as routing, DNS, database, and monitoring tools to be incorporated into the system without changing the core reasoning component.

5.1.4 Docker-Based Environment

Docker is used to create an isolated environment for the agent and target infrastructure components. The Docker network allows controlled communication between the agent server and target server.

The agent server contains the Python-based agent and infrastructure tools, while the target server provides the environment on which remote operations can be tested. SSH communication is established through port 22, while HTTP-based communication can be used for application-level testing.

This environment provides a controlled setup for testing infrastructure operations before deploying the system in a real enterprise environment.

5.2 TESTING

Testing was performed to verify the functionality of the SLM, agent, infrastructure tools, and their integration within the proposed system. Different infrastructure-related scenarios were used to examine the system's ability to interpret requests, select appropriate tools, process results, and generate responses.

TEST CASE	INPUT / CONDITION	EXPECTED RESULT	OBSERVATION
TC01 – SLM Response	Infrastructure-related user query	SLM understands the request and generates a relevant response	SLM produced an appropriate response for the given infrastructure query
TC02 – Agent Request Handling	User submits an infrastructure request	Agent interprets the request and determines the required action	Agent successfully processed the request
TC03 – Network Diagnostic	Network connectivity issue	Agent selects the appropriate network diagnostic tool	Network diagnostic tool returned the required information
TC04 – SSH Connectivity	Remote system investigation request	Agent uses SSH to communicate with the target system	SSH connectivity was successfully established
TC05 – Firewall Operation	Firewall-related request	Appropriate firewall operation is identified and executed through the tool	Firewall functionality responded to the request
TC06 – Agent Memory	Follow-up query related to a previous interaction	Agent retains and uses relevant previous context	Previous interaction context was available for the follow-up request
TC07 – Tool Result Processing	Infrastructure tool returns diagnostic information	Agent interprets the result and generates a meaningful response	Tool output was processed by the agent
TC08 – Integrated Workflow	End-to-end infrastructure request	Request passes through frontend, backend, agent, SLM, and tool layer	Components communicated successfully during testing
TC09 – Human Approval	Sensitive infrastructure operation	System requests administrator approval before execution	Approval mechanism is incorporated into the agent workflow

5.3 RESULTS AND OBSERVATIONS

The implementation and testing demonstrate the feasibility of using an SLM-based agent for AI-assisted IT infrastructure operations. The developed system establishes communication between the user interface, backend, agent, reasoning model, and infrastructure tools.

The agent is able to process infrastructure-related requests and coordinate with available tools for tasks such as network diagnostics, SSH-based investigation, and firewall

operations. Agent memory provides contextual support for follow-up interactions, while the separate tool-execution layer keeps infrastructure operations controlled.

The SLM provides the reasoning component required for understanding infrastructure-related requests and supporting the agent's decision-making process. Testing the model separately before integration provides a controlled approach for evaluating its performance.

The Docker-based environment provides an isolated setup for testing communication between the agent and target infrastructure. This enables infrastructure operations to be evaluated in a controlled environment without directly affecting production systems.

The overall implementation establishes the core foundation for an autonomous IT infrastructure operations system. The modular architecture allows additional capabilities such as RAG, DNS, routing, credential management, and Human-in-the-Loop approval to be integrated into the system.

The results demonstrate that combining an SLM, agent orchestration, memory, and controlled infrastructure tools can provide a practical foundation for assisting IT administrators with infrastructure investigation and management.

5.4 SCREENSHOTS OF IMPLEMENTATION

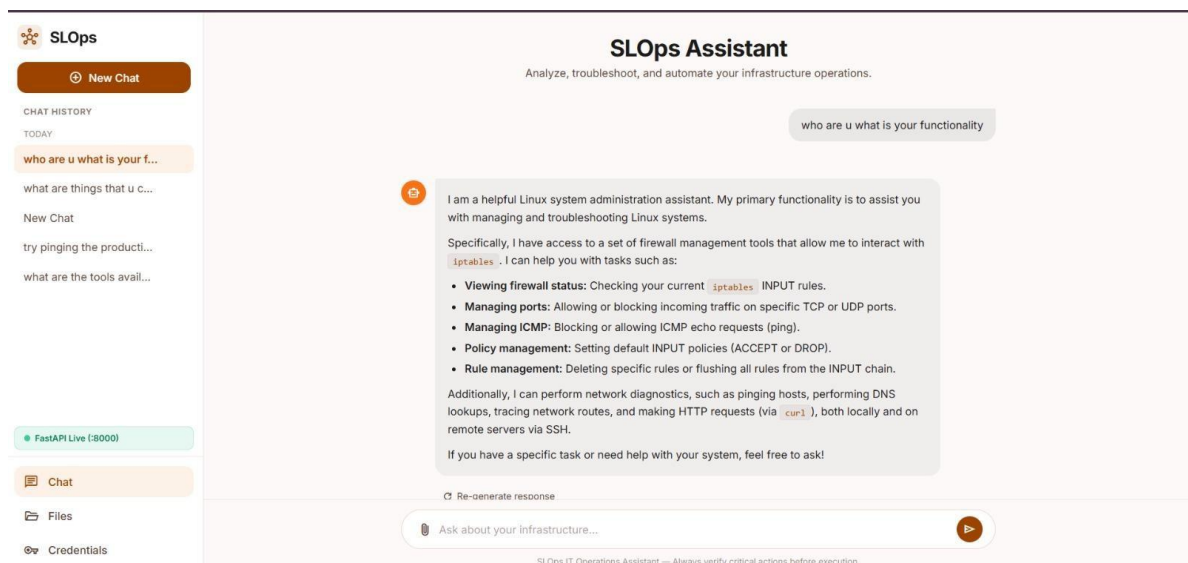


Fig. 5.1: Agent Interface and Supported Infrastructure Operations

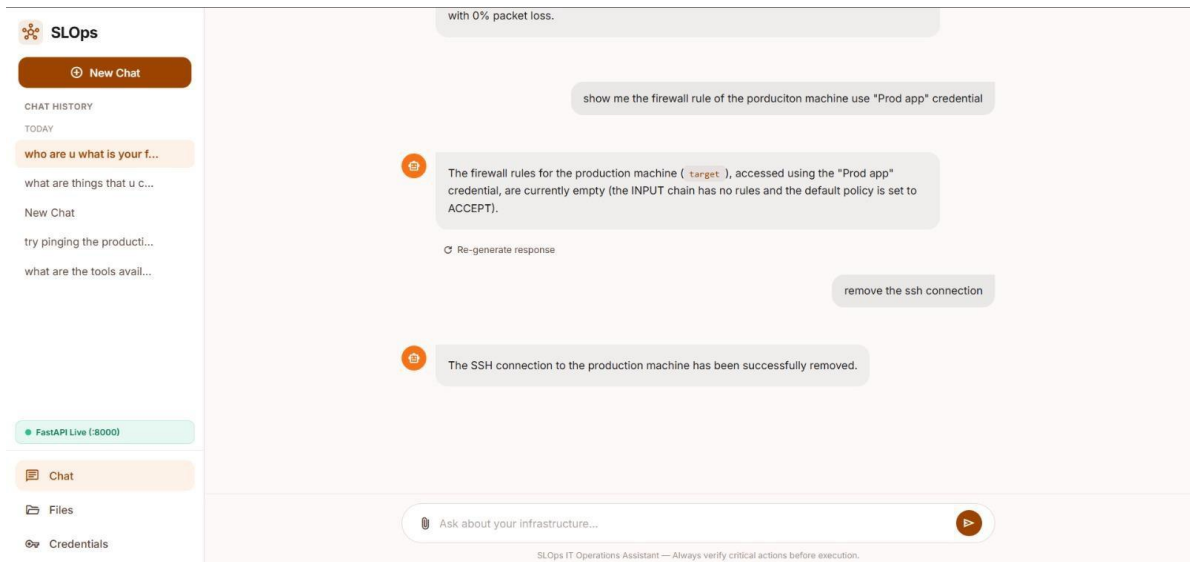


Fig. 5.2: Network Connectivity Diagnosis

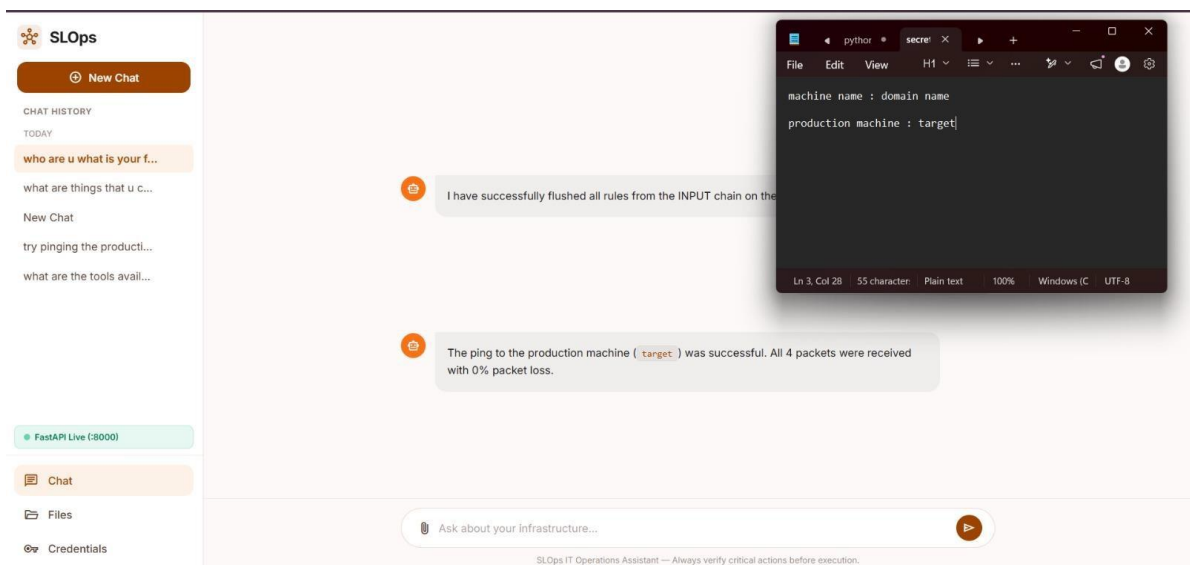


Fig. 5.3: Firewall Rule Management and SSH Operations

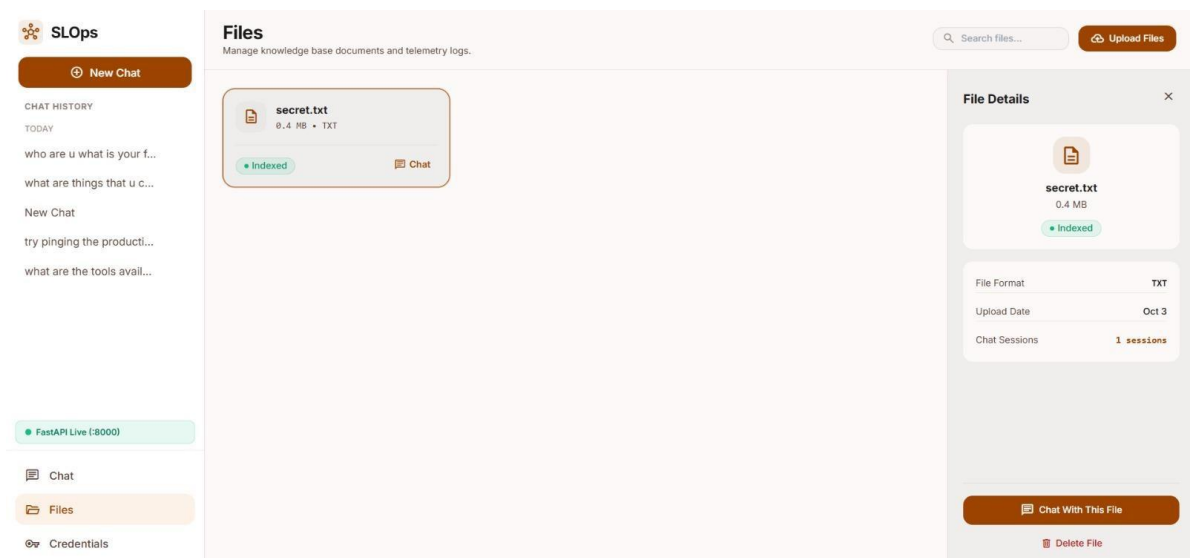


Fig. 5.4: Knowledge Base File Upload and Document Indexing

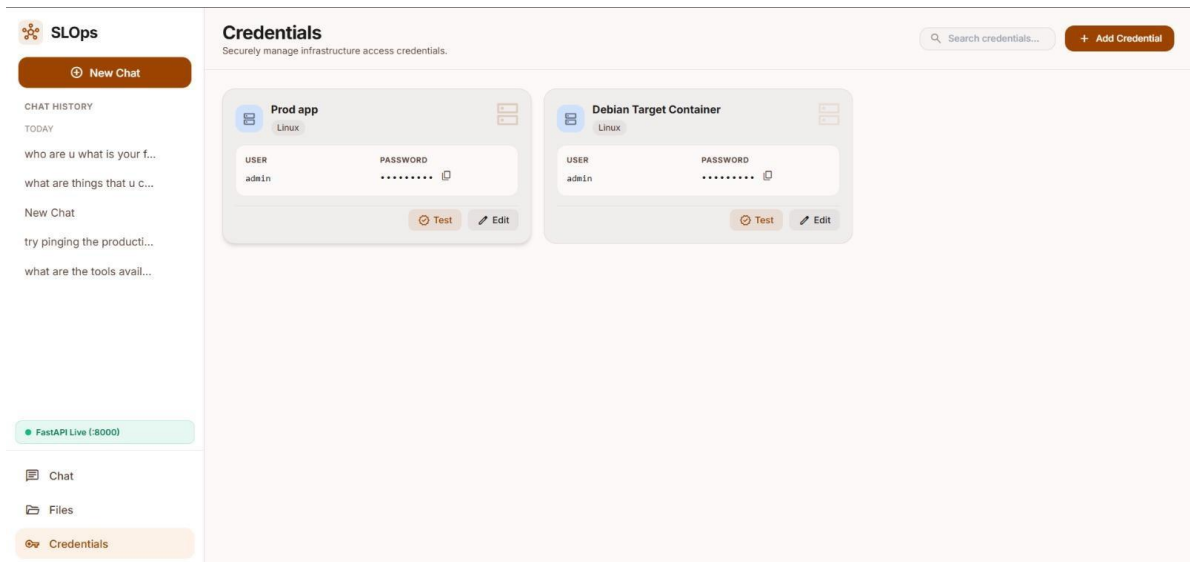


Fig. 5.5:Secure Infrastructure Credential Management

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

6.1 CONCLUSION

The proposed Autonomous IT Infrastructure Operations and Management system provides a practical foundation for AI-assisted enterprise IT operations. It addresses infrastructure complexity by introducing an intelligent layer that can interpret incidents, maintain context, and coordinate approved diagnostic capabilities.

The combination of a Small Language Model, agent orchestration, memory, Retrieval-Augmented Generation, and langchain tools provides a structured approach to connecting natural-language assistance with infrastructure operations. The local-first design emphasizes privacy and internal control, while the planned human-in-the-loop mechanism establishes administrator oversight for sensitive actions.

The current implementation demonstrates the foundational components through agents, memory, firewall functionality, basic network tools, SSH connectivity, and a frontend. Additional modules such as routing, DNS, RAG, account vault, human-in-the-loop control, database structure, and complete integration remain part of ongoing development. The project therefore establishes the architecture and core implementation required for a broader supervised IT infrastructure assistant.

6.2 FUTURE ENHANCEMENT

The following enhancements can be considered for further development of the proposed system:

- **Integrate the separately developed SLM with the Agent** to establish a complete end-to-end reasoning and decision-making workflow.
- **Expand the LangChain tool layer** to support additional infrastructure operations such as advanced routing, DNS, database, container, observability, and security operations.
- **Improve multi-agent coordination** for specialized network, Linux, database, security, and application troubleshooting.

- **Enhance evaluation and controlled automation** using larger sets of representative incidents, tool-selection accuracy, retrieval quality, response time, and safety metrics.
- **Extend enterprise-scale deployment** with stronger permission management, reliability, observability, auditability, and integration with ITSM platforms.

CHAPTER 7

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